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## **EE/EX-7005(2)-CBGS**

### **B.E. VII Semester**

Examination, December 2020

## **Choice Based Grading System (CBGS)**

### **EHVAC/DC Transmission**

**Time : Three Hours**

**Maximum Marks : 70**

**Note:** i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain the problems involved in EHV transmission.  
b) Give a neat sketch of different HVDC links.
2. a) What do you understand by "FACTS controllers"? What are the merits and demerits of FACTS controllers in power system environment?  
b) Explain the following:
  - i) IEEE-FACTS controllers
  - ii) BESS-FACTS controllers
  - iii) SMES-FACTS controllers
  - iv) STATCOM-FACTS controllers.
3. a) Explain why it is necessary to use compensation of EHVAC lines.  
b) What is SVC? Explain in brief.

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4.
  - a) Discuss the single line diagram of thyristor controlled phase shifting transformer (TC-PST). Also discuss the working principle of TC-PST.
  - b) Explain the single line diagram of thyristor controlled series capacitor (TCSC). Also discuss the working principle of TCSC.
5. Explain the following
  - a) UPFC
  - b) Explain various kinds of D.C. links.
6.
  - a) Discuss the problems associated with the harmonics introduced by HVDC converters. Also explain the characteristics of Harmonics.
  - b) Explain the following
    - i) Commutation failure
    - ii) Multi terminal D.C lines
7.
  - a) Discuss the ignition angle control scheme. Also mention its merits and demerits.
  - b) Discuss the salient features of IPEC. What are the limitations and significance of IPEC?
8. Explain the following
  - a) Over voltage in transmission system
  - b) Ignition angle control
  - c) Converter analysis Graetz circuit
  - d) Adverse effects

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